Jishnu Infotech Support Portal — Database Schema & Structure Blueprint

POSTGRESQL 18 | 28 RELATIONAL TABLES | REACT 19 + EXPRESS 5 **Production:** 129.121.128.205 • **Project:** jit_support

Project Repository: `C:\Users\HP\.gemini\antigravity\scratch\jishnu-infotech-support`

Database Engine: PostgreSQL 18.6 (Production: `129.121.128.205:5432`)

Backend: Node.js 24 + Express 5.1

Frontend: React 19 + TypeScript 5.9 + Vite 7

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1. High-Level Architecture & Entity Relationship Diagram

```
erDiagram
    COMPANIES ||--o{ USERS : "has members"
    COMPANIES ||--o{ TICKETS : "owns"
    COMPANIES ||--o{ COMPANY_JOIN_CODES : "issues"
    COMPANIES ||--o{ CLIENT_INVITATIONS : "invites"
    USERS ||--o{ TICKETS : "creates (customer_id)"
    USERS ||--o{ TICKETS : "assigned to (assignee_id)"
    USERS ||--o{ MESSAGES : "authors"
    USERS ||--o{ NOTIFICATIONS : "receives"
    USERS ||--o{ COMPANY_MEMBERSHIPS : "belongs to"
    TICKETS ||--o{ MESSAGES : "conversation"
    TICKETS ||--o{ TICKET_ATTACHMENTS : "files"
    TICKETS ||--o{ TICKET_RATINGS : "feedback (1:1)"
    TICKETS ||--o{ TICKET_ASSIGNMENT_HISTORY : "audit trail"
    TICKETS ||--o{ TICKET_TAT_EVENTS : "SLA milestones"
    USER_MANUALS ||--o{ USER_MANUAL_ATTACHMENTS : "media"
    USER_MANUALS ||--o{ USER_MANUAL_LINKS : "external links"
    USER_MANUALS ||--o{ USER_MANUAL_RECIPIENTS : "visibility"
    INFORMATION_POSTS ||--o{ INFORMATION_POST_ATTACHMENTS : "banners/docs"
    INFORMATION_POSTS ||--o{ INFORMATION_POST_COMPANY_RECIPIENTS : "target companies"
    INFORMATION_POSTS ||--o{ INFORMATION_POST_RECIPIENTS : "target clients"
    INFORMATION_POSTS ||--o{ INFORMATION_POST_NOTIFICATIONS : "read receipts"
```

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2. Complete Database Schema (All 28 Tables)

Table 1: `companies`

Primary enterprise client accounts.

Column	Type	Constraints / Default	Description
`id`	`bigint`	`PRIMARY KEY`	Auto-increment unique company ID
`name`	`text`	`NOT NULL`	Registered company trade name
`slug`	`text`	`NOT NULL UNIQUE`	URL-safe identifier
`domain`	`text`	`NULLABLE`	Company email domain (e.g. `client.com`)
`is_active`	`integer`	`DEFAULT 1 NOT NULL`	`1` = Active, `0` = Suspended
`support_valid_until`	`date`	`NULLABLE`	Annual maintenance / SLA contract expiry
`internal_consultant_id`	`bigint`	`NULLABLE`	Dedicated account manager consultant
`created_at`	`timestamp`	`DEFAULT UTC`	Registration timestamp

Table 2: `users`

Master user accounts covering all roles: Client, Company Admin, Consultant, Super Admin.

Column	Type	Constraints / Default	Description
`id`	`bigint`	`PRIMARY KEY`	User ID
`username`	`text`	`NOT NULL UNIQUE`	Case-sensitive sign-in handle
`name`	`text`	`NOT NULL`	Full legal/display name
`email`	`text`	`NULLABLE UNIQUE`	Contact email address
`contact_number`	`text`	`NULLABLE`	Phone / Mobile number
`password_hash`	`text`	`NOT NULL`	Bcrypt hashed credential
`role`	`text`	`NOT NULL`	`'client'` or `'agent'` (consultant)
`is_admin`	`integer`	`DEFAULT 0 NOT NULL`	`1` = Super Admin (Jishnu Infotech)
`is_company_admin`	`integer`	`DEFAULT 0 NOT NULL`	`1` = Company Admin (Client enterprise)
`company_id`	`bigint`	`REFERENCES companies(id)`	Primary associated company
`is_internal_consultant`	`integer`	`DEFAULT 0 NOT NULL`	Flag for dedicated internal staff
`internal_routing_priority`	`integer`	`DEFAULT 0 NOT NULL`	Weighting for auto-assignment
`routing_priority`	`integer`	`DEFAULT 0 NOT NULL`	Global round-robin priority score
`specialty`	`text`	`DEFAULT 'both'`	`'functional'`, `'technical'`, or `'both'`
`approval_status`	`text`	`DEFAULT 'approved'`	User approval workflow state
`disabled_at`	`timestamp`	`NULLABLE`	Suspension timestamp if deactivated
`created_at`	`timestamp`	`DEFAULT UTC`	Account creation date

Table 3: `company_memberships`

Enables multi-company membership so a client can switch between multiple client companies.

Column	Type	Constraints / Default	Description
`user_id`	`bigint`	`REFERENCES users(id)`	User ID
`company_id`	`bigint`	`REFERENCES companies(id)`	Company ID
`is_company_admin`	`integer`	`DEFAULT 0`	Admin rights within this specific company
`is_internal_consultant`	`integer`	`DEFAULT 0`	Dedicated consultant status for company
`internal_routing_priority`	`integer`	`DEFAULT 0`	Assignment priority for this company
`created_at`	`timestamp`	`DEFAULT UTC`	Membership granted date
Composite PK	`(user_id, company_id)`	`PRIMARY KEY`	Ensures unique user-company pair

Table 4: `tickets`

The core customer support request record (33 columns).

Column	Type	Constraints / Default	Description
`id`	`bigint`	`PRIMARY KEY`	Global unique ticket ID
`company_ticket_seq`	`integer`	`NULLABLE`	Sequential number per company (e.g. #1, #2)
`subject`	`text`	`NOT NULL`	Ticket title / summary
`description`	`text`	`NOT NULL`	Full issue report
`category`	`text`	`DEFAULT 'Other'`	`'Functional'`, `'Technical'`, `'Other'`
`priority`	`text`	`DEFAULT 'normal'`	`'low'`, `'normal'`, `'high'`, `'urgent'`
`status`	`text`	`DEFAULT 'open'`	`'open'`, `'in_progress'`, `'waiting'`, `'resolved'`, `'closed'`
`customer_id`	`bigint`	`REFERENCES users(id)`	Reporter user ID
`company_id`	`bigint`	`REFERENCES companies(id)`	Reporter company ID
`assignee_id`	`bigint`	`REFERENCES users(id)`	Assigned consultant ID
`source`	`text`	`DEFAULT 'portal'`	`'portal'`, `'email'`, or `'imported'`
`email_thread_id`	`text`	`NULLABLE`	Email Message-ID thread correlation
`first_response_at`	`timestamp`	`NULLABLE`	Timestamp of first consultant reply
`resolved_at`	`timestamp`	`NULLABLE`	Resolution timestamp
`resolved_by`	`bigint`	`REFERENCES users(id)`	Consultant who marked resolved
`reported_client_name`	`text`	`NULLABLE`	Name of contact person
`reported_client_contact`	`text`	`NULLABLE`	Phone / direct email of contact person
`historical_solution`	`text`	`NULLABLE`	Historical resolution notes
`historical_consultant_name`	`text`	`NULLABLE`	Migrated legacy consultant name
`import_key`	`text`	`NULLABLE`	Legacy migration unique ID
`tags_json`	`text`	`DEFAULT '[]'`	JSON array of ticket tags
`custom_fields_json`	`text`	`DEFAULT '{}'`	Extensible custom field key-values
`tat_start_at`	`timestamp`	`NULLABLE`	Timestamp when SLA clock began
`tat_duration_minutes`	`integer`	`NULLABLE`	SLA target window in working minutes
`tat_due_at`	`timestamp`	`NULLABLE`	Target delivery deadline datetime
`tat_elapsed_minutes`	`integer`	`DEFAULT 0`	Working minutes consumed so far
`tat_remaining_minutes`	`integer`	`NULLABLE`	Working minutes remaining before breach
`tat_status`	`text`	`NULLABLE`	`'running'`, `'paused'`, `'breached'`, `'completed'`
`tat_paused_at`	`timestamp`	`NULLABLE`	Timestamp when moved to Waiting
`tat_total_paused_minutes`	`integer`	`DEFAULT 0`	Cumulative paused duration
`tat_breached_at`	`timestamp`	`NULLABLE`	Timestamp when SLA was breached
`created_at`	`timestamp`	`DEFAULT UTC`	Created timestamp
`updated_at`	`timestamp`	`DEFAULT UTC`	Last updated timestamp

Table 5: `messages`

Threaded conversation between clients and consultants.

Column	Type	Constraints / Default	Description
`id`	`bigint`	`PRIMARY KEY`	Message ID
`ticket_id`	`bigint`	`REFERENCES tickets(id) ON DELETE CASCADE`	Parent ticket
`author_id`	`bigint`	`REFERENCES users(id)`	Author user ID
`body`	`text`	`NOT NULL`	Message content (Markdown / Plaintext)
`is_internal`	`integer`	`DEFAULT 0 NOT NULL`	`1` = **Private Consultant Note** (Hidden from clients)
`email_message_id`	`text`	`NULLABLE`	Header ID if sent or received via SMTP/IMAP
`created_at`	`timestamp`	`DEFAULT UTC`	Creation timestamp
`edited_at`	`timestamp`	`NULLABLE`	Edit timestamp if amended

Table 6: `ticket_attachments`

Files linked to tickets or individual conversation messages.

Column	Type	Constraints / Default	Description
`id`	`bigint`	`PRIMARY KEY`	Attachment ID
`ticket_id`	`bigint`	`REFERENCES tickets(id) ON DELETE CASCADE`	Target ticket
`uploaded_by`	`bigint`	`REFERENCES users(id)`	Uploader ID
`storage_name`	`text`	`NOT NULL`	Sanitized disk filename (UUID/hash)
`original_name`	`text`	`NOT NULL`	Original client filename
`mime_type`	`text`	`NOT NULL`	Validated MIME type (e.g. `application/pdf`)
`size_bytes`	`bigint`	`NOT NULL`	File size in bytes
`created_at`	`timestamp`	`DEFAULT UTC`	Upload timestamp

Table 7: `ticket_tat_events`

Audit trail of SLA and status milestone events for turnaround reporting.

Column	Type	Constraints / Default	Description
`id`	`bigint`	`PRIMARY KEY`	Event ID
`ticket_id`	`bigint`	`REFERENCES tickets(id) ON DELETE CASCADE`	Ticket reference
`event_type`	`text`	`NOT NULL`	`'created'`, `'first_response'`, `'status_change'`, `'paused'`, `'resumed'`, `'resolved'`
`status`	`text`	`NULLABLE`	Status at time of event
`old_status`	`text`	`NULLABLE`	Previous status
`new_status`	`text`	`NULLABLE`	New updated status
`event_at`	`timestamp`	`DEFAULT UTC`	Exact event timestamp
`duration_minutes`	`integer`	`DEFAULT 0`	Working minutes during this phase
`paused_minutes`	`integer`	`DEFAULT 0`	Paused minutes accumulated
`performed_by`	`bigint`	`REFERENCES users(id)`	User triggering the change
`notes`	`text`	`NULLABLE`	Reason or trigger description
`created_at`	`timestamp`	`DEFAULT UTC`	Log creation time

Table 8: `ticket_ratings`

Customer satisfaction feedback collected after ticket closure.

Column	Type	Constraints / Default	Description
`id`	`bigint`	`PRIMARY KEY`	Rating ID
`ticket_id`	`bigint`	`REFERENCES tickets(id) UNIQUE`	Guaranteed 1 rating per ticket
`customer_id`	`bigint`	`REFERENCES users(id)`	Reviewer user ID
`rating`	`integer`	`NOT NULL (1 to 5)`	CSAT star score
`comment`	`text`	`DEFAULT ''`	Client feedback review
`created_at`	`timestamp`	`DEFAULT UTC`	Submission timestamp

Table 9: `ticket_assignment_history`

Complete routing history of tickets between consultants.

Column	Type	Constraints / Default	Description
`id`	`bigint`	`PRIMARY KEY`	History ID
`ticket_id`	`bigint`	`REFERENCES tickets(id) ON DELETE CASCADE`	Ticket ID
`agent_id`	`bigint`	`REFERENCES users(id)`	Assigned consultant
`assigned_by`	`bigint`	`REFERENCES users(id)`	Assignor (or System)
`source`	`text`	`NOT NULL`	`'manual'`, `'round_robin'`, `'reassign'`
`created_at`	`timestamp`	`DEFAULT UTC`	Assignment timestamp

Tables 10–13: User Manuals & Knowledge Base

- **`user_manuals`**: `id`, `title`, `summary`, `content`, `media_kind`, `storage_name`, `original_name`, `mime_type`, `size_bytes`, `created_by`, `all_clients`, `created_at`, `updated_at`.
- **`user_manual_attachments`**: `id`, `manual_id`, `storage_name`, `original_name`, `mime_type`, `size_bytes`, `file_kind`, `created_at`.
- **`user_manual_links`**: `id`, `manual_id`, `label`, `url`, `created_at`.
- **`user_manual_recipients`**: `manual_id`, `user_id` (Composite PK).

Tables 14–18: Information Posts & Broadcast Bulletins

- **`information_posts`**: `id`, `consultant_id`, `consultant_name`, `consultant_mobile`, `information`, `all_clients`, `created_at`, `updated_at`.
- **`information_post_attachments`**: `id`, `information_id`, `uploaded_by`, `storage_name`, `original_name`, `mime_type`, `size_bytes`, `created_at`.
- **`information_post_company_recipients`**: `information_id`, `company_id`.
- **`information_post_recipients`**: `information_id`, `client_id`.
- **`information_post_notifications`**: `id`, `information_id`, `user_id`, `read_at`, `created_at`.

Tables 19–28: System Infrastructure, Authentication & Email

- **`workspace_settings`**: `key (PRIMARY KEY)`, `value`. Stores operating hours (`09:30` - `18:30`), SLA thresholds, and email routing config.
- **`holidays`**: `id`, `holiday_date`, `holiday_name`, `is_active`, `created_at`. Used to freeze TAT clock on gazetted holidays.
- **`notifications`**: `id`, `user_id`, `ticket_id`, `kind`, `message`, `read_at`, `created_at`.
- **`mail_outbox`**: `id`, `recipient`, `subject`, `body`, `message_id`, `references_header`, `status`, `attempts`, `last_error`, `created_at`, `sent_at`. Resilient background email queue.
- **`email_cursors`**: `mailbox (PRIMARY KEY)`, `uid_validity`, `last_uid`, `updated_at`. IMAP mailbox synchronization cursor.
- **`company_join_codes`**: `id`, `company_id`, `code_hash`, `created_by`, `expires_at`, `revoked_at`, `created_at`. Self-serve company registration.
- **`client_invitations`**: `id`, `company_id`, `email`, `is_company_admin`, `token_hash`, `invited_by`, `expires_at`, `accepted_at`, `created_at`.
- **`agent_invitations`**: `id`, `email`, `specialty`, `token_hash`, `invited_by`, `expires_at`, `accepted_at`, `created_at`. Consultant onboarding tokens.
- **`password_reset_tokens`**: `id`, `user_id`, `token_hash`, `expires_at`, `used_at`, `created_at`. 1-hour secure password reset tokens.
- **`schema_migrations`**: `version (PRIMARY KEY)`, `applied_at`. DDL version tracking.

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3. Project File Blueprint & Component Structure

```
C:\Users\HP\.gemini\antigravity\scratch\jishnu-infotech-support
│
├── deploy/                              # Deployment assets
│   ├── docker-compose.yml               # Multi-container orchestration
│   └── nginx.conf                       # Reverse proxy SSL config
│
├── docs/                                # Platform documentation & manuals
│   ├── BACKUP_GUIDE.md                  # Production server backup runbook
│   ├── HOW_TO_BACKUP_SERVER_DATA.html    # Printable backup guide
│   ├── HOW_TO_BACKUP_SERVER_DATA.pdf     # Rendered PDF backup manual
│   ├── MASTER_PROJECT_SPECIFICATION.md   # Architectural blueprint
│   ├── MASTER_PROJECT_SPECIFICATION.pdf  # Architectural PDF specification
│   └── PLATFORM_GUIDE.md                # End-user and administrator manual
│
├── server/                              # Node.js Express 5.1 Backend
│   ├── authz.js                         # RBAC guards & session token validator
│   ├── company-reports.js               # SLA, turnaround, and resolution metrics
│   ├── db.js                            # PostgreSQL connection pool & migrations
│   ├── email.js                         # Inbound IMAP parsing + Outbound SMTP worker
│   ├── historical-ticket-import.js      # Legacy spreadsheet importer
│   ├── index.js                         # Main Express application entry point
│   ├── internal-routing.js              # Consultant round-robin & workload routing
│   ├── tat-engine.js                    # Turnaround time & business-hours SLA engine
│   ├── ticket-numbering.js              # Sequential ticket number generator
│   └── validation.js                    # MIME whitelist, sanitize, and input checks
│
├── src/                                 # React 19 + TypeScript Frontend
│   ├── App.tsx                          # Master client router & portal shell
│   ├── index.css                        # Tailwind / CSS layout styling
│   ├── main.tsx                         # React DOM mount point
│   │
│   ├── components/                      # Shared UI Widgets
│   │   ├── AttachmentViewer.tsx         # PDF, docx, and image preview modal
│   │   ├── Layout.tsx                   # Dual-portal sidebar & top header navigation
│   │   ├── NotificationDropdown.tsx     # Real-time bell notification drawer
│   │   ├── StatusBadge.tsx              # Colorful status & priority pills
│   │   ├── TATIndicator.tsx             # SLA countdown & breach status badge
│   │   └── TicketRatingModal.tsx        # 5-star customer feedback modal
│   │
│   ├── context/                         # State Management Contexts
│   │   ├── AuthContext.tsx              # Sign-in state, active role, active company
│   │   └── NotificationContext.tsx      # Live alert polling state
│   │
│   └── pages/                           # Portal Screen Views
│       ├── ClientInbox.tsx              # Company tickets & personal ticket queue
│       ├── ConsultantInbox.tsx          # Shared queue, Assigned to me, filters
│       ├── CreateTicket.tsx             # Multi-file ticket submission form
│       ├── CustomerDirectory.tsx        # Client enterprise & staff management
│       ├── KnowledgeBase.tsx            # User manuals & documentation browser
│       ├── InformationBulletin.tsx      # Broadcast announcements & notices
│       ├── Reports.tsx                  # SLA analytics, TAT averages, CSV/PDF export
│       ├── Settings.tsx                 # Working hours, holidays, routing config
│       └── TicketDetail.tsx             # Conversation thread, internal notes, actions
│
```

```
├── package.json                    # Dependencies & NPM scripts
├── tsconfig.json                   # TypeScript compiler configuration
└── vite.config.ts                  # Vite compilation & API proxy config
```

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4. Key Functional Engines

1. The SLA Turnaround Time (TAT) Engine (`server/tat-engine.js`)

* **Operating Hours:** Default 09:30 AM to 06:30 PM (9 hours/business day).
* **Calendar Exclusions:** Skips Saturdays, Sundays, and dates matching `public.holidays`.
* **State Behavior:**
* **Open / In Progress:** Clock actively accumulates `tat_elapsed_minutes`.
* **Waiting for Client:** Clock pauses (`tat_paused_at`). The paused minutes are added to `tat_total_paused_minutes`.
* **Resolved / Closed:** Clock halts and logs a final `resolved` milestone in `ticket_tat_events`.

2. Dual-Channel Ticket Ingestion

1. **Web Portal:** Authenticated submission via `POST /api/tickets` with Multer validation (PDF, JPG, PNG, WebP up to 10MB).
2. **Email Ingestion:** `imapflow` checks incoming emails, correlates with existing tickets via `email_thread_id` / `References` headers, or creates a new ticket assigned by default routing.

3. Outbox Email Worker (`server/email.js`)

Emails are never sent synchronously inside request handlers. Instead, handlers write a record to `mail_outbox`. A resilient background worker picks up `pending` items, attempts delivery via SMTP, records failures in `last_error`, and implements exponential backoff.